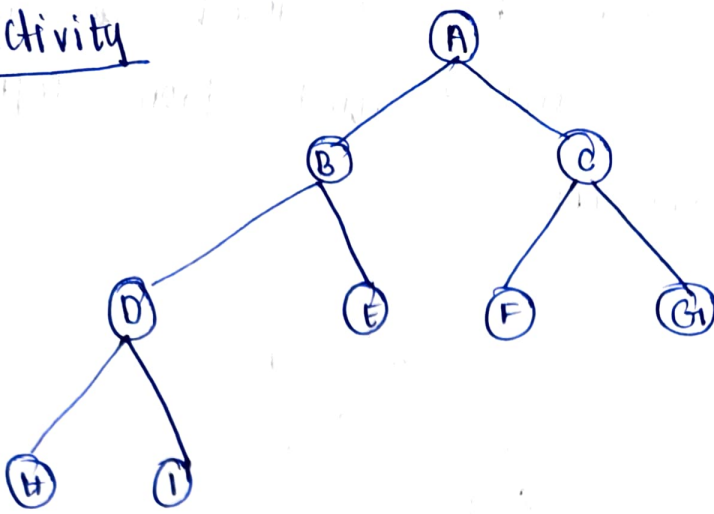
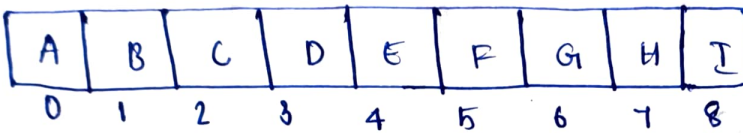


Activity



Case (i):



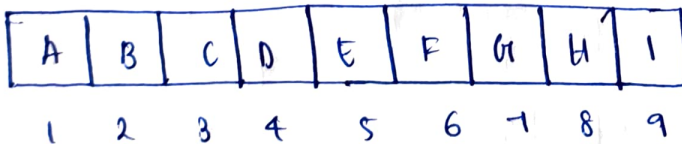
$$\text{parent} = \frac{(i-1)}{2}$$

$i=4$ eg

$$\text{left child} = (2 \times p) + 1$$

$$\text{right child} = (2 \times p) + 2$$

Case (ii)



$$\text{parent} = \frac{i}{2}$$

$$\text{left child node} = 2 \times p$$

$$\text{right} = (2 \times i) + 1$$

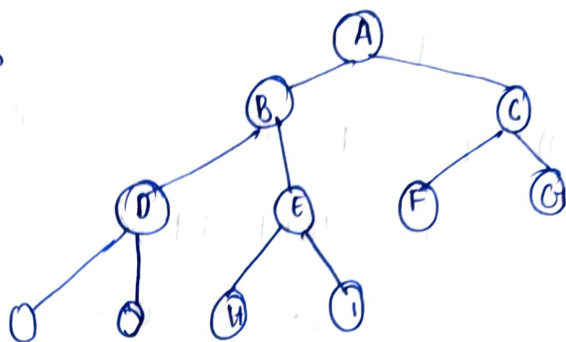
for $i = 4$

case (i)

$$pn = \frac{4-1}{2} = 1.5 = 1 \Rightarrow B$$

$$lch = (2 \times 4) + 1 = 9 \propto$$

$$rch = (2 \times 4) + 2 = 10 \propto$$



case (ii)

$$pn = \frac{4}{2} = 2 = B$$

$$lch = 4 \times 2 = 8 = H$$

$$rch = (4 \times 2) + 1 = 9 = I$$

if the node or not

right make it right.

for $i = 7$

case (i) =

$$pn = \frac{7-1}{2} = 3 = D$$

$$lch = (2 \times 7) + 1 = 15 \propto$$

$$rch = (2 \times 7) + 2 = 16 \propto$$



should be in

binary ~~not~~ ~~istatuku~~.

case (ii)

$$pn = \frac{7}{2} = 3.5 = 3 = C$$

$$lch = 2 \times 7 = 14 \propto$$

$$rch = (2 \times 7) + 1 = 15 \propto$$

A	B	C	D	E	F	G	-	-	H	I
0	1	2	3	4	5	6	7	8	9	10

$$i = 9$$

$$\text{case (i)} \Rightarrow pn = 4 = E$$

$$len = 18 + 1 = 19 \times$$

$$rcn = 18 + 2 = 20 \times$$

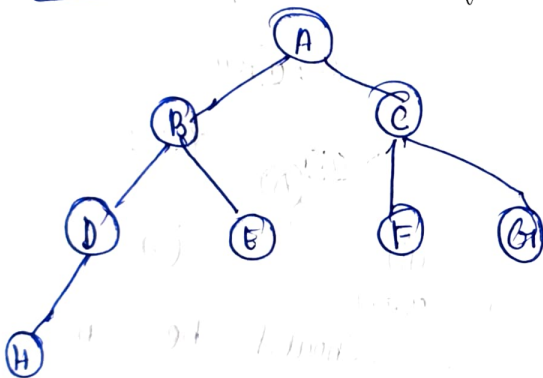
$$\text{case (ii)} \Rightarrow pn = 4.5 = 4 = D$$

$$len = 18 = \times$$

$$rcn = 18 + 1 = \times$$

Home task

eg-3



1	2	3	4	5	6	7	8
A	B	C	D	E	F	G	H
0	1	2	3	4	5	6	7

$i = 5$

$$pn = \frac{4}{2} = 2 = C$$

$$len = 10 + 1 = 11 = \times$$

$$rcn = 10 + 2 = 12 = \times$$

$$pn = \frac{5}{2} = 2.5 = 2 = B$$

$$len = 10 = \times$$

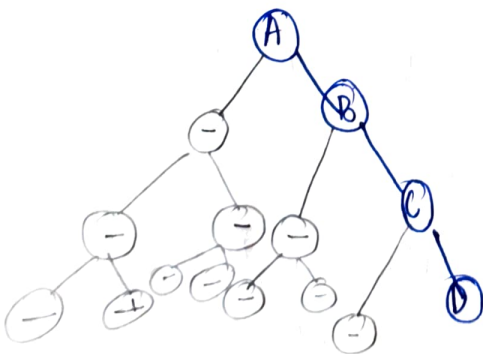
$$rcn = 11 = \times$$

The index value

changes when adding

empty node.

eg-4



$$i = 3$$

1	2	3	4	5	6	7	8	9	10	11	12	13
A	-	B	-	-	-	C	-	-	-	-	-	-
0	1	2	3	4	5	6	7	8	9	10	11	12

$i = 3$

$$pn = \frac{2}{2} = 1 = A$$

$$len = 6 + 1 = 7 = C$$

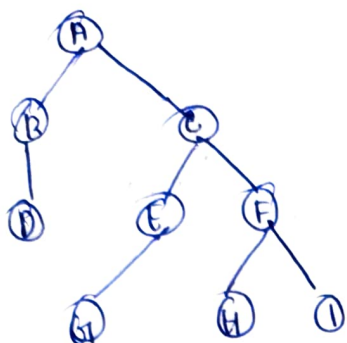
$$rcn = 6 + 2 = 8 = \times$$

$$pn = \frac{3}{2} = 1.5 = 1 = A$$

$$len = 6 = \times$$

$$rcn = 6 + 1 = 7 = C$$

Binary tree traversal



In Order - left Root Right

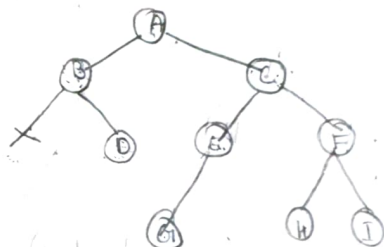
pre order - Root Left Right

post order - left Right Root

In order - BDAGECHF I

pre " - ABDCEGH FHI

post " - DBG EHI FCA

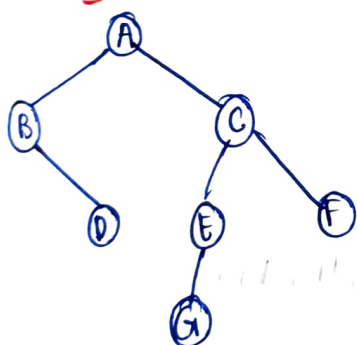


BDAGECHF I

ABDCEGH FHI

DBG EHI FCA

Home task



In - BDAGECHF

pre - ABDCEGH F

post - DBG EFC A

traversal from left ~~Root~~ node (A) first left subtree is (B) node it is a root so move left, no lcn print root node (B) then print (D) - ~~then~~ now left subtree is completed. now print Root

(A) Backtrack

now traversal right subtree in same way as B D A